

# Nebiyou Hailemariam

412-251-7954 | [nebhailema@gmail.com](mailto:nebhailema@gmail.com) | [linkedin.com/in/nebhailemariam](https://www.linkedin.com/in/nebhailemariam) | [nebhailemariam.github.io/](https://github.com/nebhailemariam)

## EDUCATION

### Carnegie Mellon University — CGPA 3.93

Aug. 2022 – May 2024

*Masters in Information Technology, Applied Machine Learning Specialization*

### Addis Ababa University — CGPA 3.8

Sep. 2016 – May. 2020

*Bachelor of Science, Computer Science*

## EXPERIENCE

### Software Engineer II, AI

Aug. 2024 - Present;

*Motive*

*Remote*

- Building and maintaining a **multi-tenant car dealership platform** used by 300+ dealerships across the U.S. and Canada, serving millions of users, leveraging **Python (FastAPI)**, **Ruby on Rails**, and **React**.
- Developed an **agentic conversational system** using **LangChain**, **LangSmith**, **FastAPI**, and **Pydantic**, capable of creating and updating webpages, performing content analytics, and generating SEO-optimized blog posts, enabling dealership admins to scale content automation.
- Fine-tuned **lightweight reranker models** using **PyTorch**, **Sentence Transformers**, **Hugging face**, and **Vertex Workbench** to train a **content-based filtering recommendation system** to improve search result relevance and boost conversion rates.
- Set up **training and deployment pipelines** on **GCP Vertex AI** (Cloud Storage, Artifact Registry, Vertex Training, Vertex Model, Vertex Endpoint) and used **Weights & Biases** for experiment tracking and monitoring.
- Utilized the **pytest** framework to test AI microservice functionality, structuring tests with the **Arrange-Act-Assert** pattern for clarity and maintainability.

### Software Engineer I, AI

Aug. 2024 - March 2025;

*eezly*

*Remote*

- Worked on **eezly**, a grocery price comparison application used by over 30,000+ users, leveraging **ASP.NET Core Web API**, **Python (FastAPI)**, **PyTorch**, and cloud-based microservices to build scalable, AI-driven app.
- Built a **Recipe Recommendation System** using **LangChain**, **OpenAI**, and the **Recipe1M+** dataset, creating a **Retrieval-Augmented Generation (RAG)** system to suggest recipes based on the products users purchase. Incorporated the **Weaviate** vector database to enhance search and recommendation.
- Employed **PyTorch** and **Hugging Face** to train hierarchical machine-learning models for classifying retail products from various stores (e.g., Walmart) into **aisles, categories, and subcategories**.
- Integrated **Gorse**, a recommender system, and contributed to open-source recommender systems.
- Designed and implemented **RESTful APIs** for inventory management using n-tier architecture and developed a **single-page application** with **React.js**.
- Implemented **OAuth 2.0 client-credential flow** using **OpenIddict** for secure machine-to-machine communication, **Single Sign-On (Firebase, Cognito)**, and **ASP.NET Core Identity** for user management.
- Designed a messaging system using **Kafka** with **Golang** as the message producer.
- Employed **xUnit** and **pytest** to write unit and integration tests for microservices using a custom web application factory.

### Software Engineer I

May 2019 - Jul. 2020

*Matrix Information Technology*

*Remote*

- Developed scalable microservices for **eQub**, a platform used by **tens of thousands of users**, transforming a traditional financial communal saving system into a modern **digital solution**.
- Implemented key features like **transaction processing** and **account management**, leveraging **ASP.NET Core Identity**, **Entity Framework**, and **AWS RDS**.
- Developed microservices applications for the **no-code data and operations management platform WorkOnGrid**, a platform used by **thousands of users**, leveraging **AWS (EC2, RDS, EKS, SES, SQS)**.
- Built and deployed **predictive models** for the **user dashboard** of the **no-code platform WorkOnGrid**, enhancing users' **decision-making processes**.
- Led the development of a **subscription management microservice** integrated with **Stripe**, simplifying subscription workflows and boosting **customer satisfaction**.

- Wrote comprehensive **unit and integration tests** for **ASP.NET Core Web API** using **xUnit** and for **Python** using **pytest**, significantly improving **code reliability** and reducing bugs across services.

## Software Engineer I

Aug. 2020 – Sep. 2021

Digital Medarbeider

Oslo, Norway

- Developed a **Cloud-Based School Management System** for managing students, employees, and schedules, enhancing administrative efficiency and automation for **Norway municipalities**.
- Implemented a **microservices architecture** for student and employee management, and scheduling systems
- Employed **IdentityServer4** for securing microservices and **ASP.NET Core Identity** for user authentication.
- Leveraged **Azure App Service** for web hosting, **Azure DevOps** for CI/CD pipelines, and **Azure Key Vault**.
- Utilized various databases, including **Azure Cosmos DB** for scalable NoSQL data storage, and **Azure SQL Database** for relational data needs, ensuring **efficient and reliable data handling**.
- Wrote **unit and integration tests**, used **Bogus** for realistic test data, and documented APIs with **Swagger** to facilitate integration and maintenance.

## Research Assistant in Machine Learning

May 2021 – Jan 2022

Empathic Computing Lab

Auckland, New Zealand

- **Empathic Computing Laboratory (ECL)** is an academic research laboratory directed by **Prof. Mark Billingham** at the **University of South Australia** in Adelaide, Australia, and the **University of Auckland** in Auckland, New Zealand.
- Collaborated with **Ph.D. students** to refine methods for detecting **emotions from physiological signals**.
- Conducted extensive **literature reviews** and analyzed the performance of various **machine learning and deep learning models**, applying rigorous Hyperparameter tuning. Authored a **14-page paper (IUI - ACM)**.

## PROJECTS

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### NebTorch — a minimal Autograd engine built from scratch using NumPy

Sep 2023 – Dec 2023

- NebTorch is a minimal **Autograd engine** built from scratch using **NumPy**, inspired by PyTorch's automatic differentiation system.
- After completing and serving as a TA for **11-785: Introduction to Deep Learning** at CMU taught by Prof. Bhiksha Raj, I was inspired to build my own Autograd engine from scratch.
- Building NebTorch has been very rewarding—I've solidified my understanding of **Deep Learning** and **Automatic Differentiation**, and gained appreciation for frameworks such as **PyTorch** and **TensorFlow**.

### XLM-R Based Extractive Amharic Question Answering with AmaSQuAD

Sep 2023 – May 2024

- Completed my thesis project at Carnegie Mellon University in multilingual question-answering research (NLP).
- Developed a novel framework for translating the SQuAD 2.0 dataset into Amharic, resulting in the creation of an open-source dataset called AmaSQuAD.
- Implemented a translation-based data generation framework valuable for extractive Question Answering (QA) systems, contributing to the advancement of natural language processing (NLP) for low-resource languages.
- Leveraged XLM-R, a pre-trained language model, and fine-tuned it specifically for Amharic QA tasks, achieving 7% F1 improvements in baseline performance.
- The dataset is publicly available on [Hugging Face](#).

### Machine Learning Based Rain Gauge Using Acoustic Data

Sep 2022 – May 2023

- Addressed the challenges of conventional weather stations such as high setup costs, instrument fragility, and measurement errors by exploring alternative sound sources for making rainfall predictions.
- Used a Convolutional Neural Network (CNN) regression model using PyTorch and TensorFlow to estimate rainfall intensities from MFCCs extracted from acoustic recordings. Employed CNN model and achieved a Mean Absolute Percentage Error (MAPE) of 35.20% and a Mean Squared Error (MSE) of 0.66, outperforming a baseline Support Vector Regression (SVR) model with 152.55% MAPE and 1.73 MSE.

## TECHNICAL SKILLS

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**Languages:** Python, C#, Go, JavaScript, Java

**Frameworks:** ASP.NET Core Web API, Entity Framework, IdentityServer4, OpenIddict, Dependency Injection, Flask, FastAPI, Django, Gin, Express.js, Node.js, React, Node.js, PyTorch, TensorFlow, Scikit-learn, Pandas, Numpy, RabbitMQ, Kafka

**Dev Tools:** Git, Docker, Kubernetes, GCP, AWS, Azure, Postman, Jenkins, Grafana, Prometheus, Grafana-Loki, Jaeger

**Web & Frontend Technologies:** TypeScript, ReactJS, JavaScript, CSS, Bootstrap, Redux

**Cloud Services & Platforms:** AWS (EC2, EKS, RDS, SES, SQS, MQTT), Google Cloud (Google Compute Engine, Google Cloud Storage, Google Colab), Firebase

**Security:** OAuth 2.0, JWT, IdentityServer4, ASP.NET Core Identity, Passport.js

**Databases & Data Storage:** Relational (MySQL, PostgreSQL, Postgres), NoSQL (MongoDB, DynamoDB, Cosmos DB, Redis), Vector (Weaviate), ORM (SQLAlchemy, Entity Framework)

**Testing & Quality Assurance:** Unit test, Integration test, xUnit, WebApplicationFactory, Bogus, pytest, Mocha, Jest

**DevOps & CI/CD:** Docker, Kubernetes, Jenkins, Containerization

**Machine Learning & Deep Learning Skills:** Linear Regression, Support Vector Machines, Bagging and Boosting, Neural Networks, CNNs, RNNs, LSTMs, NLP, Graph Neural Networks (GNNs), clustering, Graph Attention Networks (GATs), Hyperparameter Tuning, Transfer Learning, TensorFlow, PyTorch, Scikit-learn, XGBoost, Hugging Face Transformers, Weights & Biases, Computer Vision, LangChain

**Data Science:** Data Pipelines (Kafka, RabbitMQ, MQTT), Data Visualization (matplotlib), Data Analysis, Scikit-learn, Pandas, NumPy, PyTorch, NLTK

**Architectures:** Event-Driven, Microservices, Serverless, Monolithic Architecture, Test-Driven Development (TDD), Design patterns, Agile, REST API